

Social Security

Danny Yagan
UC Berkeley
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Poll (No Right Answer)

Why do we have Social Security?

- A. Market failure (adverse selection in the annuity market)
- B. Individual failure (myopia/self-control problems)
- C. Other/Uncertain

Poll (Partial Right Answer)

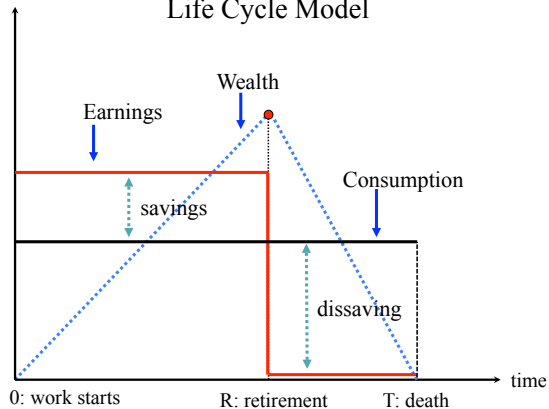
What is a key drawback to Social Security?

- A. Moral hazard
- B. One-size-fits-all savings rate
- C. Illiquidity

Retirement Problem

- Life-Cycle: Individuals ability to work declines with aging but individuals continue to live after they are unwilling/unable to work
- Standard Life-Cycle Model Prediction: Absent any government program, rational individual would save while working to consume savings while retired
- Optimal saving problem is extremely complex: uncertainty in returns to saving, in life-span, in future ability/opportunities to work, in future tastes/health
- In practice: When govt was small before 20th century $\Rightarrow$ Many people worked till unable to (often till close to death) and then were taken care of by family members [US elderly poverty rate very high before Social Security]

Life Cycle Model



Cumulative American Retirees by Age

2017 Federal Reserve Survey of Household Economics and Decisionmaking

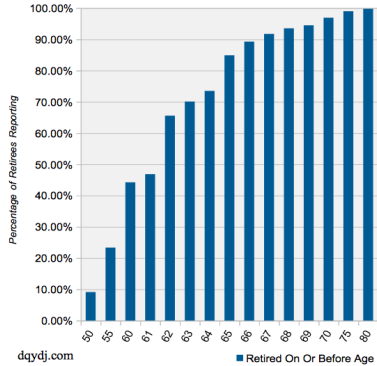
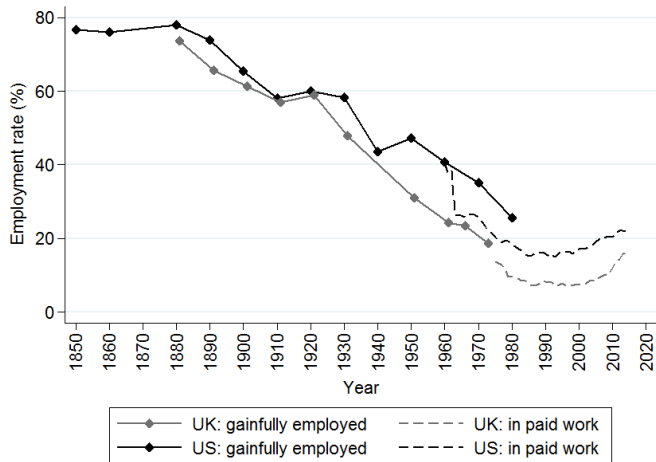
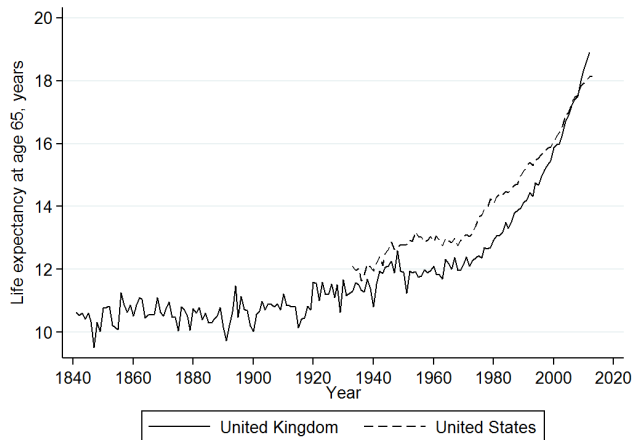


Figure 2.6: Employment rate of men aged 65+ in the UK and the US



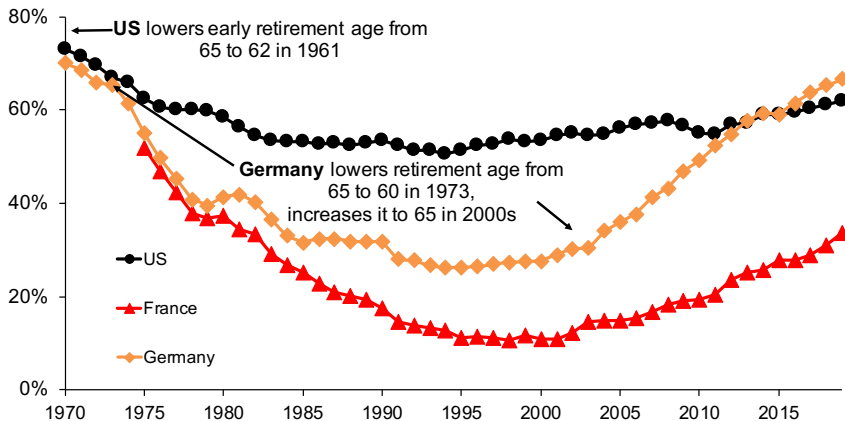
Source: Blundell, French, and Tetlow (2017)

Figure 2.7: Life expectancy of men at age 65 in the UK and the US



Source: UK data from the Office for National Statistics, US data from the Human Mortality Database.
Source: Blundell, French, and Tetlow (2017)

Panel B. Employment rates of men aged 60-64, 1970-2019



Source: Saez '21 using OECD database

Government Intervention in Retirement Policy

- Actual Retirement Programs: All OECD countries implement substantial government funded retirement programs (substantial share of GDP around 6-10%, US smaller around 5%), started in first part of 20th century and have been growing.
- Common structure: Individuals pay social security contributions (payroll taxes) while working and receive retirement benefits when they stop working till the end of their life (annuity)
- Extension of the earlier family model: it's no longer your own working kids who take care of you in old age but all workers in the country
- In the United States, the public retirement program is called Social Security

Social Security: Program Details

- How Is Social Security Financed?
 - Almost all workers in the United States pay the Federal Insurance Contributions Act (FICA) tax on their earnings.
 - Tax is 12.4% of earnings (6.2% paid by employer, 6.2% paid by employees) up to a cap of \$184,500 (in 2026)
- Who Is Eligible to Receive Social Security?
 - A person must have worked and paid this payroll tax for 40 quarters (10 years) over their lifetime, and must be of age 62 or older.

- How Are Social Security Benefits Calculated?
 - Annuity payment: A payment that lasts until the recipient's death.
 - The amount of this annuity payment is a function of the recipient's average (taxable) earnings over the person's 35 highest earning years where each month's earnings are expressed in today's dollars (AIME = average indexed monthly earnings)
 - Once benefits start for a given person, they are indexed to price inflation once every year ("real" annuity)

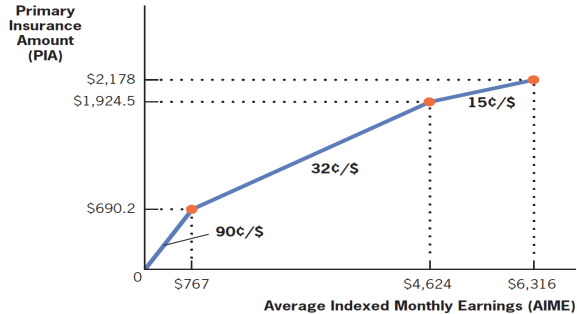
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APPLICATION: Why Choose 35 Years?

- Using the 35 highest years reflects multiple concerns.
 - No penalty for low-earning years early in career.
 - Not too large a benefit for high earning years late in career.
- Too short a window leads to abuse:
 - Bus driver working 25-hour shifts to maximize pension payment.
 - Brazilian public employees receiving promotions right before retirement.

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Social Security Benefits as a Function of Earnings



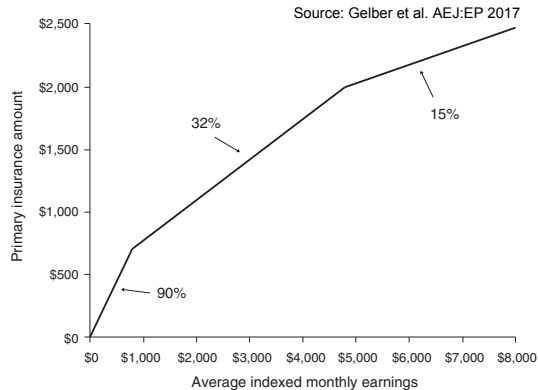


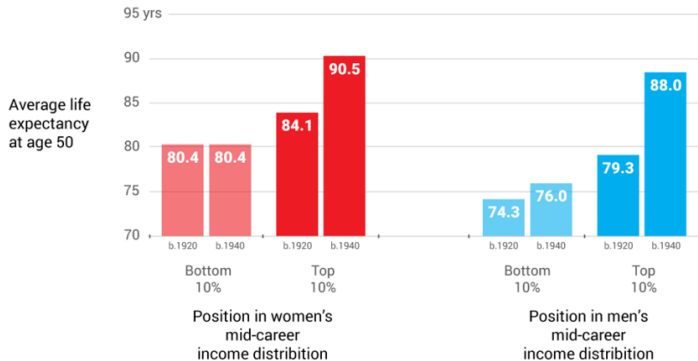
FIGURE 1. PRIMARY INSURANCE AMOUNT AS A FUNCTION OF AVERAGE INDEXED MONTHLY EARNINGS

Notes: The figure shows the primary insurance amount (PIA) as a function of average indexed monthly earnings (AIME) in 2013. The percentages are marginal replacement rates.

Source: SSA (2013)

Americans making more money are living longer than those earning less

This means gaps in life expectancy by income have grown over time.



Source: Bosworth et al. 2016

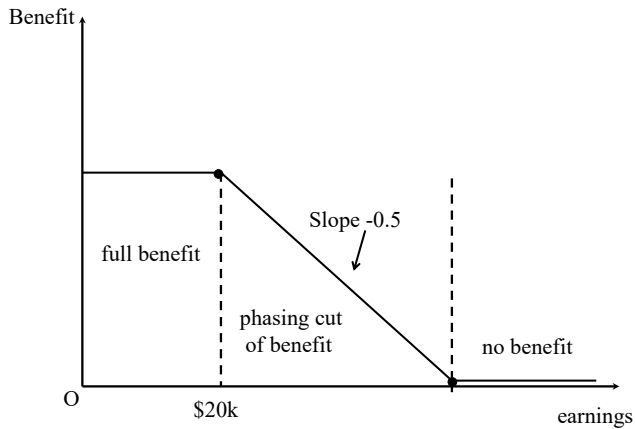
How Are Social Security Benefits Paid Out?

- Full Benefits Age (FBA): The age at which a Social Security recipient receives full retirement benefits (Primary Insurance Amount): currently 67 if born 1960+ (used to be 65)
- Early Entitlement Age (EEA): The earliest age at which a Social Security recipient can receive reduced benefits: currently 62
- If you claim benefits 1 year before FBA, you get 8% less in annual benefits (permanently), if you claim 2 years before FBA, you get 16% less in annual benefits, etc.
- You get 8% more in benefits if you claim 1 year after FBA. Benefits automatically paid at 70.

Social Security: Program Details

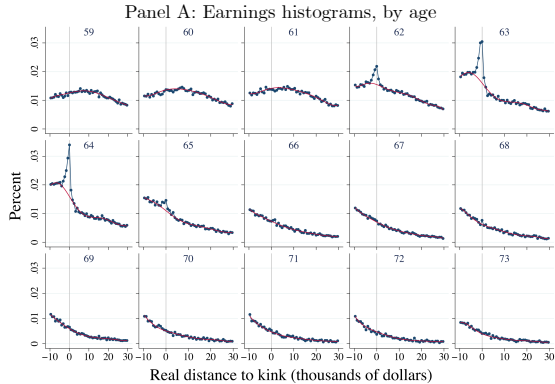
- Can You Work and Receive Social Security?
 - The *earnings test* reduces the benefits of 62 to 66-year olds by \$0.50 for each dollar of earnings they have above about \$24K
 - Not really a tax because later benefits are increased (as if you had retired later) but most people don't understand the system
- Are There Benefits for Family Members?
 - Spouses of claimants (get own benefits or 50% of primary earner benefits, whichever is bigger)
 - Children of deceased workers
 - Spouses who survive a Social Security recipient

Earning test for Social Security Benefit



Source: Gelber, Jones, Sacks (2013)

Figure E.6: Adjustment Across Ages: Histograms of Earnings and Normalized Excess Mass, 59-73-year-olds Claiming OASI by Age 65, 2000-2006



Sources of Retirement Income

- Govt provided retirement benefits (US Social Security): For 2/3 of US retirees, SS is more than 50% of income. 1/3 of elderly households depend almost entirely (90%+) on SS.
- Home ownership: 79% of US elderly are homeowners
- Employer pensions (tax favored): about 50% of elderly US households have employer pension benefits. Two types:
 - Traditional: defined benefit and mandatory — employer carries full risk [in sharp decline, many in default]
 - New: defined contribution and elective: 401(k)s — employee carries full risk
- Extra savings through non-tax favored instruments: significant only for wealthy minority (roughly 10% of retirees)

Funded vs. Unfunded Programs

- Two forms of retirement programs:
 - ① Unfunded (pay-as-you-go): benefits of current retirees are paid out of contributions from current workers [generational link]. Current benefits = current contributions
 - ② Funded: workers contributions are invested in financial assets and will pay for benefits when they retire [no generational link]. Current benefits = past contributions + market returns on past contributions
- Social security (as most public retirement systems) is unfunded
- Most private pension plans (such as 401(k)s) are funded

Funded vs. Unfunded Systems

- Funded system: each generation gets a market return r on contributions: $\text{benefits} = \text{tax you paid} \cdot (1 + r)$
- Unfunded system: 1st generation of retirees gets free benefits when the system starts
 - For later generations: pay tax (for older generation) and you get benefits from younger generation
 - Implicit return on taxes is the sum of population growth n and real wage growth (per worker) g
 - $\text{Benefits} = \text{tax paid} \cdot (1 + n)(1 + g) \simeq \text{tax paid} \cdot (1 + n + g)$

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What Is Social Security and How Does It Work?

How Does Social Security Work Over Time?

How Social Security Redistributes Income

■ TABLE 13-1

Social Security in a Two-Period World

Period	Number of Young Workers	Earnings Per Young Worker	Taxes Paid Per Young Worker	Total Taxes Paid	Number of Old Retirees	Benefits to Old Retirees	Taxes Paid by Old Retirees	Rate of Return
1	100	\$20,000	0	0	0	0	—	—
2	105	\$21,000	\$2,100	\$220,500	100	\$2,205	0	Infinite
3	110	\$22,050	\$2,205	\$242,550	105	\$2,310	\$2,100	10%
4	115	\$23,153	\$2,315	\$266,225	110	\$2,420	\$2,205	10%
5	121	\$24,310	0	0	115	0	\$2,315	Negatively infinite

Funded vs. Unfunded Systems

- Unfunded system is always desirable when $n + g > r$ (Diamond 1965): an economy with $n + g > r$ is called dynamically inefficient and introducing an unfunded system makes a Pareto improvement
- US economy: Annual $n = 1\%$ and $g = 1\%$ [$n + g$ was higher in 1940-1970]. $r \simeq 5\%$. In general $r > n + g$ in practice.
- Note that r is much more risky than $n + g$: risk adjusted market rate of return should be lower than average market rate r but still higher than $n + g$
- Funded system delivers higher returns because it does not deliver a free lunch to 1st generation
- Choice between funded vs. unfunded system is an inter-generational redistribution trade-off

How Does Social Security Redistribute in Practice?

- Social Security Wealth (SSW): The expected present discounted value of a person's future Social Security payments minus the expected present discounted value of a person's payroll tax payments.
- SSW is computed as follows:
 - Calculate the entire future stream of benefits that a person expects to receive before he or she dies.
 - Use a discount rate to calculate the present discounted value (PDV) of that stream of benefits.
 - Calculate the entire future stream of social security taxes that a person expects to pay before he or she dies.
 - Compute the PDV of that stream of taxes.
 - Take the difference between these two to get the SSW.

13.1

How Does Social Security Redistribute in Practice? SSW for a Single Male

Earnings	Turns 65 in 1960	Turns 65 in 1995	Turns 65 in 2030
Low earner	\$26,100	\$12,500	-\$4,100
Average earner	35,500	-5,100	-56,200
High earner	35,800	-41,100	-248,500

- Redistribution from younger to older cohorts due to:
 - First cohort didn't pay in until 1937.
 - Payroll tax has increased over time.

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How Does Social Security Redistribute in Practice?

Some examples of how SSW varies within groups that are the same ages include the following:

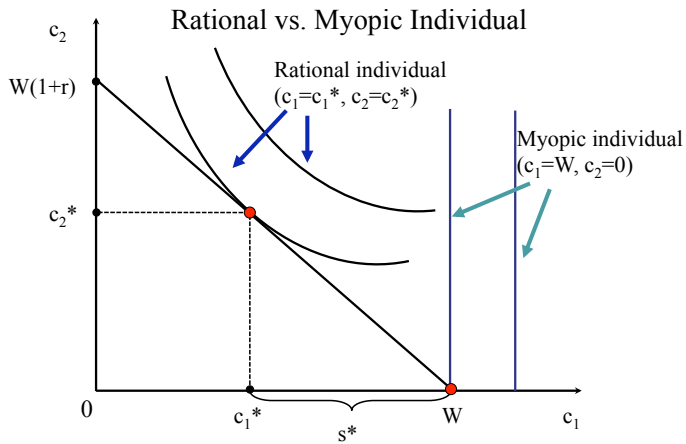
- Females have more SSW than males because they live longer.
- Married couples have more SSW than single people.
- Single-earner couples have more SSW than two-earner couples.
- The gains to the poor relative to the rich from Social Security are overstated because the length of life rises with income.

Rationales for Social Security

- A. Individual Failure
 - Without a public program, people won't save enough for their own retirement because of myopia, self-control problems, information (how much to save, how to invest savings)
 - Popularity of Social Security suggests that people understand their own failures and the need for government intervention
- B. Adverse selection in the annuities market
 - The longer a person lives, the less money the insurer makes from an annuity contract
 - $\Rightarrow$ People with short life expectancy less likely to buy annuities
 - This could lead to such a high price for annuities that most potential buyers would not want to buy them

Model: Rational vs. Myopic Savers

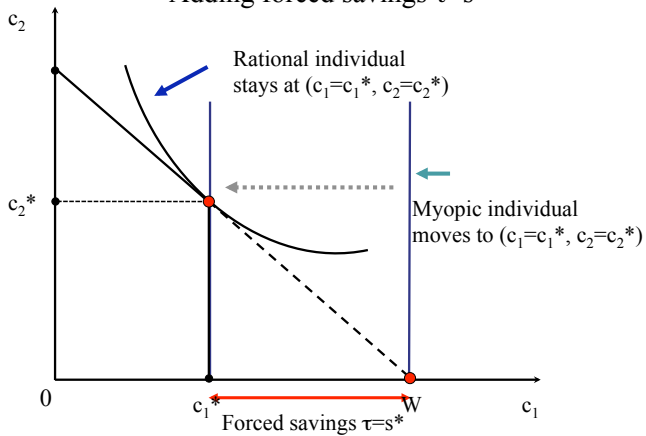
- Rational individuals:
 - $\max u(c_1) + \delta u(c_2)$ st $c_1 + s = w$ and $c_2 = s \cdot (1 + r)$
 - $\Rightarrow c_1 + c_2/(1 + r) = w$
 - FOC: $u'(c_2)/u'(c_1) = 1/[(1 + r)\delta]$, let s^* be optimal saving
 - Example: If $\delta = 1$ and $r = 0$ then $s^* = w/2$ and $c_1 = c_2 = w/2$
- Myopic individuals:
 - $\max u(c_1)$ st $c_1 + s = w$ and $c_2 = s \cdot (1 + r) \Rightarrow c_1 = w$ and $s = c_2 = 0$



Model: Rational vs. Myopic Savers

- Social welfare is always $u(c_1) + \delta u(c_2)$
- Govt imposes forced saving tax τ such that $\tau = s^*$ and benefits $b = \tau \cdot (1 + r)$. Cannot borrow against b [as in current Social Security]
- Rational individual unaffected: adjusts s one-to-one so that outcome unchanged [rational unaffected as long as $\tau \leq s^*$]: 100% crowding out of private savings by forced savings
- Myopic individual affected (0% crowding out): new outcome maximizes Social Welfare
- Forced savings is a good solution: does not affect those responsible, affects the myopic individuals in the socially desired way

Adding forced savings $\tau=s^*$



- Universal forced savings vs. means-tested program
- An alternative to forced savings is to just have a means-tested program for poor elderly (who did not save) and financed by tax on everybody
- 2 drawbacks of means-tested program:
 - Transfer from responsible individuals to myopic individuals
 - Incentives to under-save to get means-tested pension
- $\Rightarrow$ Forced savings program does not penalize responsible people and is likely to be more stable politically

Crowd-Out Effect of Social Security on Savings

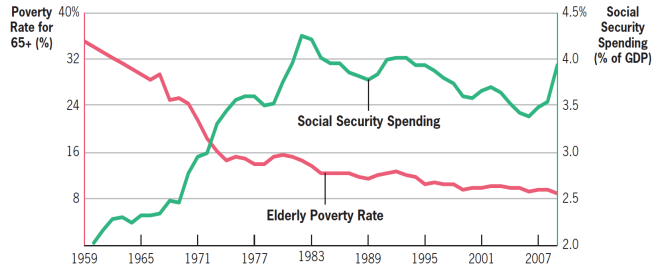
- The effect of Social Security on private savings has been the subject of a large number of studies over the past 30 years
- To measure the impact of Social Security on savings, there must be a way to compare people with different levels of Social Security benefits who are otherwise identical
- In the United States, Social Security is a national program that applies to almost all workers; very similar people usually have very similar benefits. Recent studies have provided evidence on the impact of Social Security-like programs on private savings in Italy.
- Italian Reforms in 1992 substantially reduced the benefits, and thus future SSW, for younger workers in the public sector, while reducing much less the benefits of older workers and those in the private sector.
- Studies estimate that about 30–40% of the reduction in SSW was offset by higher private savings.

Evidence for Myopia and Adequate Savings

- Diamond JpubE 1977: old age poverty has fallen as Social Security expanded. Poverty for other groups has not fallen nearly as much
- Fall in consumption during retirement: Hamermesh (1984) shows that consumption falls by 5% per year for the elderly [consumption is not smooth but not necessarily suboptimal]
- Fall in consumption at retirement: Bernheim, Skinner, Weinberg (2001) show that drop in consumption is significant for all groups except the wealthiest [consistent with myopia]

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Living Standards of the Elderly, 1959–2009



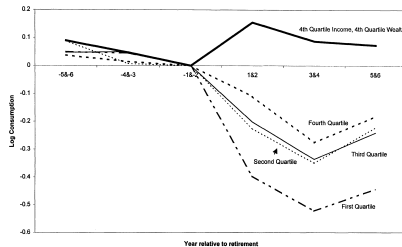


FIGURE 4. CHANGE IN CONSUMPTION AT RETIREMENT, BY WEALTH QUARTILE

Source: Bernheim et al. (2001), p. 847

Consumption Drop at Retirement: Aguiar-Hurst JPE05

- Starting point: Empirically, consumption falls with retirement...but studies use expenditures as measure of consumption
- Aguiar-Hurst JPE05 show that it is important to differentiate between consumption and expenditures. Further, the paper provides new information on the complementarity of consumption and leisure after retirement.
 - ① Confirm that expenditure on food falls by 17% at retirement but
 - ② Time spent on home production rises by 60%
 - ③ All measures of caloric intake, vitamin intake, meat quality, etc. do not drop at retirement (though strikingly find that caloric intake falls when getting unemployed)

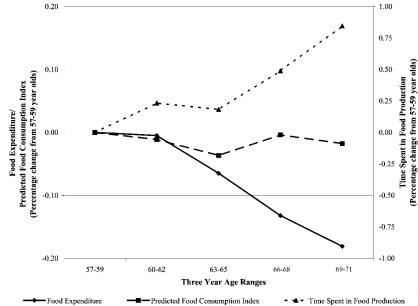


FIG. 1.—Percentage change in food expenditure, predicted food consumption index, and time spent on food production for male household heads by three-year age ranges. Data are taken from the pooled 1989–91 and 1994–96 cross sections of the CSFII, excluding the oversample of low-income households. The sample is restricted to male household heads (1,510 households). All series were normalized by the average levels for household heads aged 57–59. All subsequent years are the percentage deviations from the age 57–59 levels. See Sec. IV for details of data and derivation of food consumption index

Source: Aguiar and Hurst (2005), p. 925

Social Security and Retirement: Theory

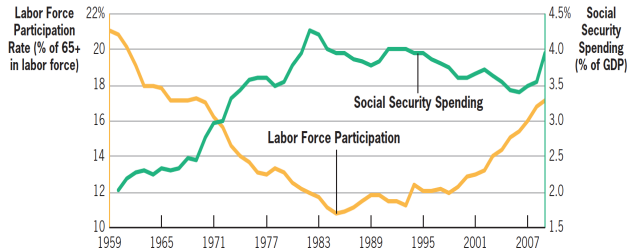
- If a 62-year-old worker works until 63, instead of retiring at 62 and claiming her Social Security benefits, three things happen through the Social Security system:
 - ① She pays an extra year of payroll taxes on her earnings.
 - ② She receives one year less of Social Security benefits.
 - ③ She gets a higher Social Security benefit level through the actuarial adjustment (8% extra permanently per year of delay)
- Adjustment is called actuarially fair if those 3 effects cancel out in PDV (US system has been reformed to be close to fair on average)

Social Security and Retirement: Theory

- Two key elements of a social security system may affect retirement behavior:
 - ① Availability of benefits at Early Retirement (Entitlement) Age (EEA): (62 in US). Those effects arise because of myopia or lack of information [a rational individual is not affected by EEA because he/she can use own savings while retired till he/she reaches age 62]
 - ② Non-actuarially fair adjustments of benefits for those retiring after the EEA: If benefits are not adjusted in a fair way, they can create a huge implicit tax on work (US used to have very little adjustment, naturally not based on mortality rate)

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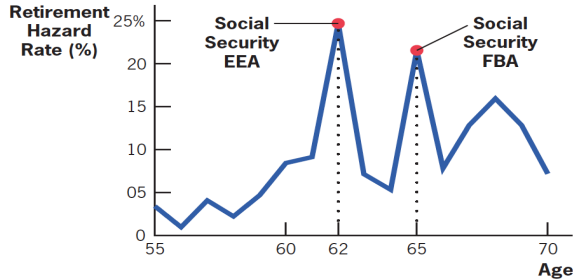
Elderly Work and Social Security, 1959–2009



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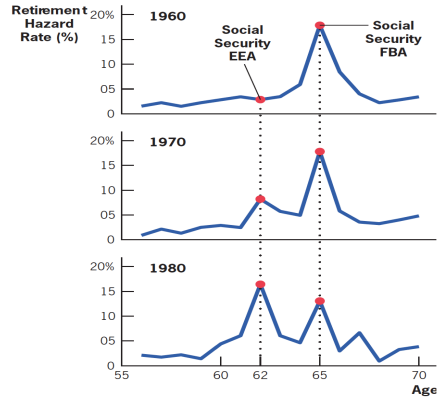
Spike in Retirement Hazard at EEA

- **Retirement hazard rate:** The percentage of workers retiring at a certain age.



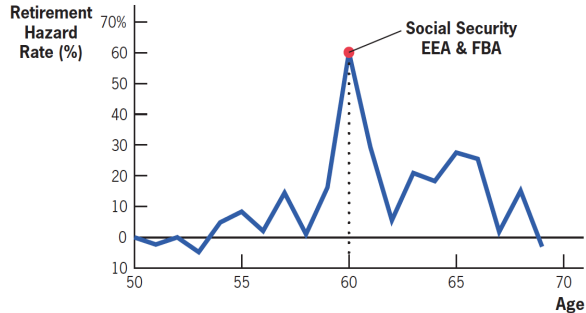
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Spike in Retirement Hazard at EEA



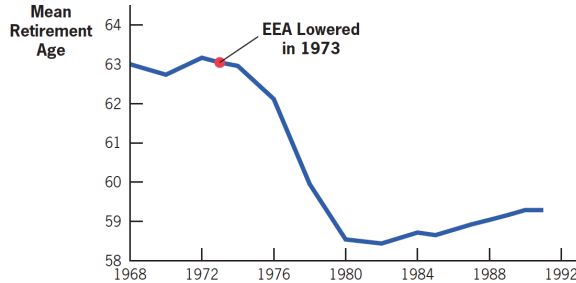
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Retirement Hazard Rate in France



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Evidence: Retirement Age in Germany, 1968–1992



- Retirement age lowered from 65 to 60 in 1973.

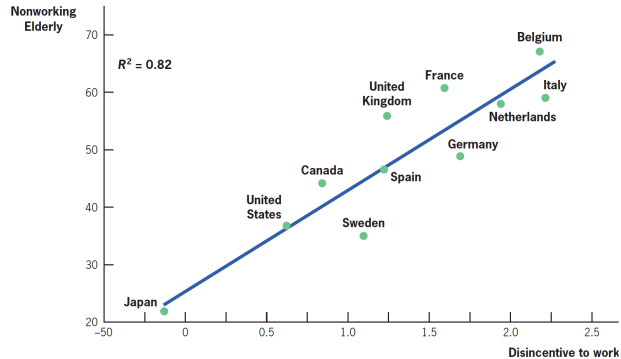
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APPLICATION: Implicit Social Security Taxes and Retirement Behavior

- Gruber and Wise (1999) calculated the implicit tax from Social Security for a series of countries.
- Across countries, there is a great deal of variation in the implicit tax rate.
 - Implicit tax close to zero for 62-year-olds in the United States.
 - 91% in the Netherlands.
- And countries with higher taxes have less elderly labor force participation.

13.3

Implicit Social Security Taxes and Retirement Behavior



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Social Security and Retirement: Implications

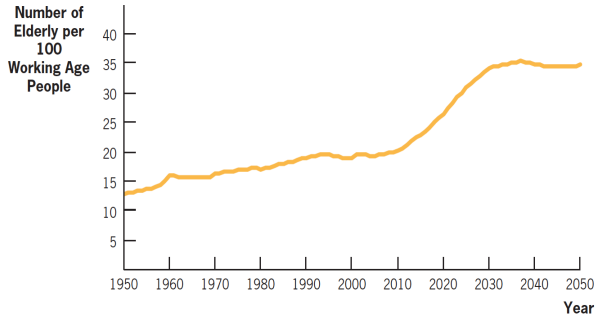
- Evidence suggests that it is potentially very costly to design Social Security systems that allow very early retirement and/or penalize additional work beyond the retirement age.
- Adjusting systems to more fairly reward work at old ages can mitigate much of the efficiency cost of Social Security
- Alternative: have an early retirement age that is not too low and provide disability benefits to those who truly cannot work and haven't yet reached the early retirement age

Social Security Reform: Problems with Current System

- Rate of return $n + g$ has declined from over 3% to about 2% due to:
 - ① n : Retirement of baby boom large cohorts born 1945-1965
 - ② Increase in life expectancy at retirement age (top half has seen large gains while bottom half has stagnated in recent decades)
 - ③ g : Slower productivity growth since 1975 (from 2% to 1%)
- 1)+2) imply number of elderly per working age person increases from .15 in 1960 to .35 in 2030
- Solvency requires adjusting taxes or benefits

13.4

Social Security Reform



1983 Greenspan Commission

- Demographic changes are predictable, so 1st reform was implemented in 1983 (designed to solve budget problems over next 75 years)
 - ① Increased payroll taxes to build a trust-fund
 - ② Increased retirement age in the future (from age 65 to 67)
- Trust fund invested in Treasury Bills (Fed gov debt):
$$TF_{t+1} = TF_t \cdot (1 + i) + SSTax_t - SSBen_t$$
- Trust fund peaked at \$2.8 Tr in 2013, will be exhausted by 2032, taxes will then cover about 75% of promised benefits

13.4

APPLICATION: The Social Security Trust Fund and National Savings

- In theory, one benefit of the partial funding of Social Security through the build-up of the trust fund is an increase in national savings.
- The trust fund is “off budget,” not supposed to be part of budget discussion.
- But typically the government reports the deficit/surplus from the “unified budget,” which incorporates off-budget categories.
- Makes it easy to treat trust fund as an asset, avoid fixing the deficit.

Social Security Reform Options

- Increase contributions: Democratic presidential candidate consensus is to apply Social Security taxes to high earnings (e.g. above \$250,000 or \$400,000). Democratic alternative is broad-based payroll tax increase (Larson House plan).
- Reduce benefits for all: Straight cut not politically likely but there are proposals to:
 - Index retirement age to life expectancy
 - Index benefits to chained-CPI instead of CPI after retirement
 - Make benefits fully taxable for income tax
- Reduce benefits for rich (i.e. means-tested benefits): could substantially reduce private savings if based on wealth in retirement rather than working-life income. Could also undermine broad-based support.
- Invest Trust Fund in higher yield assets (such as stock-market, as proposed by Clinton in 1990s): No longer an option since now depleting Trust Fund.
- Give up: Privatization

Social Security Privatization

- Two components:
 - ① Funding the system
 - ② Create individual accounts (like private employer 401k pensions). Current benefits = past contributions + market return
- Controversial academic and policy debate, a number of countries have privatized their social security systems (Chile, Mexico, UK)
- Main proponent: Feldstein (Harvard), Main critic: Diamond (MIT)
- Pro argument: get higher return on contributions $r > n + g$, increase capital stock and future wages.

Social Security Privatization Accounting

- Exactly the reverse of pay-as-you-go calculations:
 - ① First generation loses as they need to fund current retirees and own contributions. All future generations gain [generational redistribution]
 - ② If govt increases debt to pay for current retirees: future generations get higher return on contributions but need to re-pay higher govt debt $\Rightarrow$ Complete wash for all generations
- $\Rightarrow$ Only way funding generates real changes is by hurting some transitional generations which have to double pay

Additional Privatization Issues

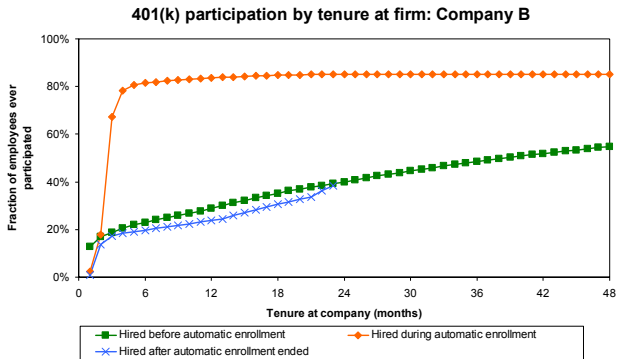
- Risk: individuals bear investment risk (stock market fluctuates too much relative to economy) and cannot count on defined level of benefits [Privatization would need to include minimum pension provision]
- Annuitization: hard to impose in privatized system because of political constraints [hard to force sick person to annuitize her wealth] $\Rightarrow$ Some people will exhaust benefits before death and be poor in very old age [looming problem with 401(k)s]
- Investment mistakes: Investing for retirement is a hard problem, and predatory financial advisers make it harder.
- Administrative costs: privatized systems (Chile, UK) admin costs very high (1% of assets) due to wasteful advertisement by mutual funds

Evidence on Investment Mistakes

- 401(k) private pensions in the US offer evidence of difficult-to-rationalize investment behavior
 - ① Default effects: opt-in vs. opt-out have enormous effects on 401(k) enrollment [Madrian and Shea QJE'01]
 - ② $1/N$ investment choices of 401(k) contributions: many people invest contributions by dividing them equally into investment options (regardless of the options)
 - ③ People often invest 401(k) in company stock which is extremely risky (Enron). Strong evidence of default effects in investment choices as well
 - ④ Evidence that financial education and advice has impacts on savings decisions (Thaler and Benartzi JPE '04: Saving More Tomorrow experiment).
- Can explain existence of Social Security

Automatic enrollment effect

Automatic enrollment dramatically increases participation.

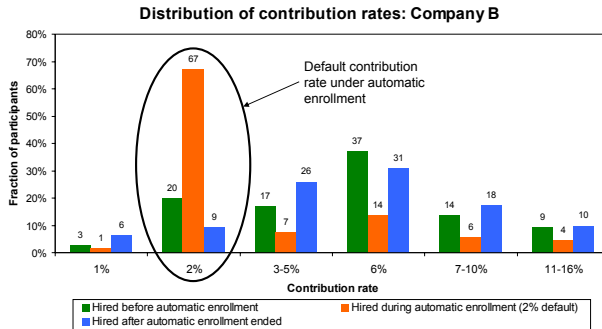


Source: Madrian and Shea (2001)

6

Automatic enrollment effect

Employees enrolled under automatic enrollment cluster at the default contribution rate.



Source: Madrian and Shea (2001)

7

The Flypaper Effect in Individual Investor Asset Allocation (Choi, Laibson, Madrian 2007)

Studied a firm that used several different match systems in their 401(k) plan.

I'll discuss two of those regimes today:

Match allocated to employer stock and workers can reallocate

- Call this “default” case (default is employer stock)

Match allocated to an asset actively chosen by workers; workers *required* to make an active designation.

- Call this “no default” case (workers must choose)

Economically, these two systems are identical.

They both allow workers to do whatever the worker wants.

Source: courtesy of David Laibson

Consequences of the two regimes

	<u>Balances in employer stock</u>	
	<u>Default ES</u>	<u>No Default</u>
Own Balance in Employer Stock	24%	20%
Matching Balance in Employer Stock	94%	27%
Total Balance in Employer Stock	56%	22%

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Source: courtesy of David Laibson

- Social Security is the largest social insurance program in the United States, and the largest single expenditure item of the federal government
- Key reasons for existence of social security programs: likely undersaving by some individuals for retirement, though there is a traditional adverse selection rationale as well
- Social Security faces a long-run financing problem requiring to increase taxes or cut benefits in the long-run
- The question of how to resolve this problem will be one of the most contentious sources of political debate over the next twenty years

Poll (No Right Answer)

Why do we have Social Security?

- A. Market failure (adverse selection in the annuity market)
- B. Individual failure (myopia/self-control problems)
- C. Other/Uncertain

Poll (Partial Right Answer)

What is a key drawback to Social Security?

- A. Moral hazard
- B. One-size-fits-all savings rate
- C. Illiquidity

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